WORKED SOLUTION

STEP I 2007, Question 1

Pure • Counting Digit Arrangements

> **STEP I 2007, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/07-s1-q1

The key idea

Don't try to list balanced numbers. Instead, sort them by the common value s of the two digit sums. For each s, the first pair of digits and the last pair can be chosen independently, so the number of balanced numbers with that sum is a product of two counts. Everything then comes down to counting the ways of writing s as the sum of two digits, which is the same as counting lattice points on a diagonal of a square grid.

Solution

Write a 4-digit balanced number as $abcd$, where each digit lies in $\{0, 1, \ldots, k\}$, $a \geq 1$ (the leading digit can't be 0), and

$$a + b = c + d = s$$

for some s with $0 \leq s \leq 2k$. For each s let

- $N(s)$ be the number of ordered pairs (c, d) of digits with $c + d = s$;
- $M(s)$ be the number of ordered pairs (a, b) of digits with $a \geq 1$ and $a + b = s$.

Once s is fixed, any admissible front pair can go with any admissible back pair, so the number of balanced numbers with digit sum s is $M(s)N(s)$. Different values of s give different numbers, so

$$\text{total} = \sum_{s=0}^{2k} M(s)\, N(s).$$

Counting the pairs

$N(s)$. If $0 \leq s \leq k$, then c can be any of $0, 1, \ldots, s$, and $d = s - c$ is automatically a digit. So $N(s) = s + 1$. If $k < s \leq 2k$, then $d = s - c \leq k$ forces $c \geq s - k$, so c runs from $s - k$ to k, and $N(s) = 2k + 1 - s$.

$M(s)$. This is the same count, except that we lose the pair with $a = 0$, if it exists. That pair is $(0, s)$, and it is allowed only when $s \le k$. So

$$M(s) = \begin{cases} s, & 0 \le s \le k, \\ 2k + 1 - s, & k < s \le 2k. \end{cases}$$

In particular $M(0) = 0$: no balanced number has digit sum 0, which is right, since $a \ge 1$.

Ask yourself Why can't we just count all four-digit strings with $a + b = c + d$ and leave it there?

Because strings such as 0312 aren't 4-digit numbers. The condition $a \ge 1$ affects only the front pair, and only for small sums, which is exactly why $M(s)$ and $N(s)$ differ just when $s \le k$.

Part (i)

Here $k = 4$, and the sums run from 0 to 8:

s	0	1	2	3	4	5	6	7	8
$M(s)$	0	1	2	3	4	4	3	2	1
$N(s)$	1	2	3	4	5	4	3	2	1
$M(s)N(s)$	0	2	6	12	20	16	9	4	1

The bottom row adds up to $(2 + 6 + 12 + 20) + (16 + 9 + 4 + 1) = 40 + 30 = 70$. So seventy 4-digit balanced numbers can be made from the digits 0 to 4, as required.

Part (ii)

Now split the sum at $s = k$, using the formulae for M and N:

$$\text{total} = \sum_{s=1}^{k} s(s+1) \; + \sum_{s=k+1}^{2k} (2k + 1 - s)^2.$$

(The $s = 0$ term is zero.) In the second sum put $r = 2k + 1 - s$. As s runs from $k + 1$ to $2k$, r runs from k down to 1, so the second sum is $\sum_{r=1}^{k} r^2$. Hence

$$\begin{aligned} \text{total} &= \sum_{r=1}^{k} (r^2 + r) + \sum_{r=1}^{k} r^2 = 2\sum_{r=1}^{k} r^2 + \sum_{r=1}^{k} r \\ &= \tfrac{1}{3}k(k+1)(2k+1) + \tfrac{1}{2}k(k+1) \\ &= \tfrac{1}{6}k(k+1)\big(2(2k+1) + 3\big) \\ &= \tfrac{1}{6}k(k+1)(4k+5), \end{aligned}$$

using the given identity for $\sum r^2$ (the $r = 0$ term contributes nothing) and $\sum_{r=1}^{k} r = \frac{1}{2}k(k+1)$. This is the required result.

Checks

- Putting $k = 4$ into the formula gives $\frac{1}{6} \cdot 4 \cdot 5 \cdot 21 = 70$, agreeing with part (i).
- For $k = 1$ the formula gives 3. Directly, the 4-digit balanced numbers using only 0 and 1 are 1001, 1010 and 1111.
- A brute-force count over all strings $abcd$ with $a \geq 1$ matches the formula for every k from 1 to 15 (for example 615 when $k = 9$, the full set of decimal digits).

Common slips

- Forgetting the leading-digit condition, which gives $\sum N(s)^2 = \frac{1}{3}(k+1)(2k^2+4k+3)$ instead, which is 85 when $k = 4$.
- Getting the number of pairs wrong for $s > k$: it is $2k + 1 - s$, not $s + 1$, because both digits are capped at k.
- Counting the middle sum $s = k$ in both halves of the split, or in neither.

Beyond STEP

The count $N(s)$ is the coefficient of x^s in $(1 + x + \cdots + x^k)^2$, and the total number of balanced numbers is a coefficient of a product of such polynomials. This *generating function* method handles $2n$-digit balanced numbers too, where direct counting quickly gets out of hand. It is a central tool of undergraduate combinatorics. For more counting problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

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